

for loop:

It is an entry control loop.

for is a keyword.

It is also used to repeat a program several times based on a condition.

When compared with while and do while, for loop is looking to be smart. In for it is compulsory to maintain two semicolons.

For works without condition also and default condition is always 1 i.e. true.

Generally for loop is having 3 expressions.

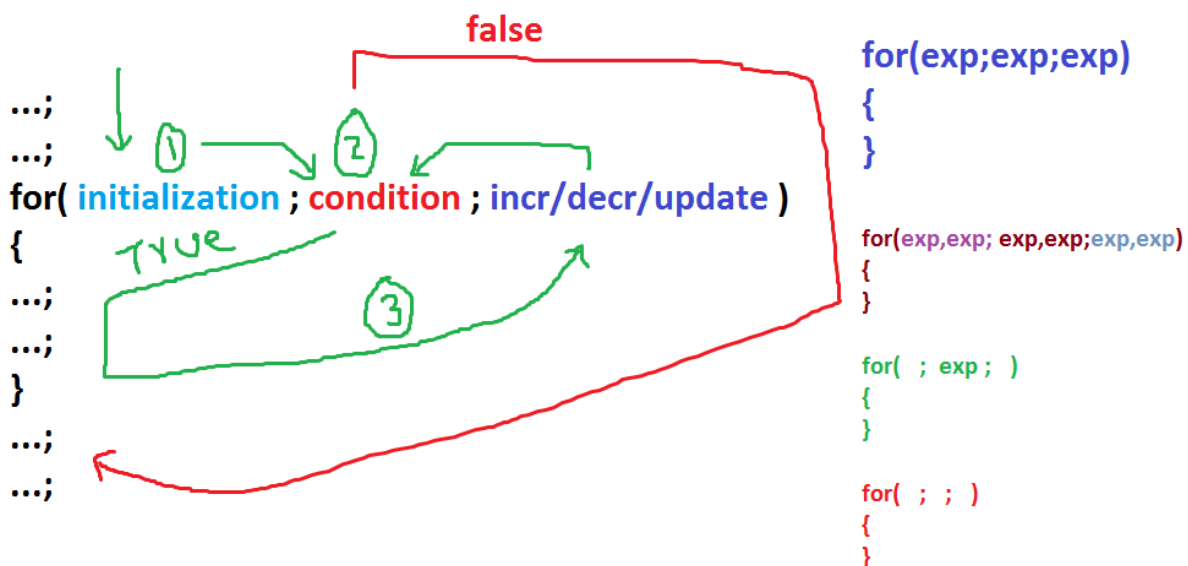
- 1. Initialization**
- 2. Test condition / expression**
- 3. Increment/decrement / updation**

At first entry of for loop the initialization part is executed and later the test condition is checked. If the condition is true then the for block statements are executed. After completion of the block, the increment or decrement part is executed. Later once again the test condition is evaluated. If it is true then once again for block statements are executed. Like this the process is continued until the condition becomes false. Here the initialization part is executed only once, at the time of loop beginning.

It is mandatory to maintain 2 semicolon (;) in a for loop.

If the for loop is having more than three expressions, it is mandatory to separate the expressions with , separator.

If the for loop is having less than three expressions, then leave the expressions with empty semicolon.



Printing the given table?

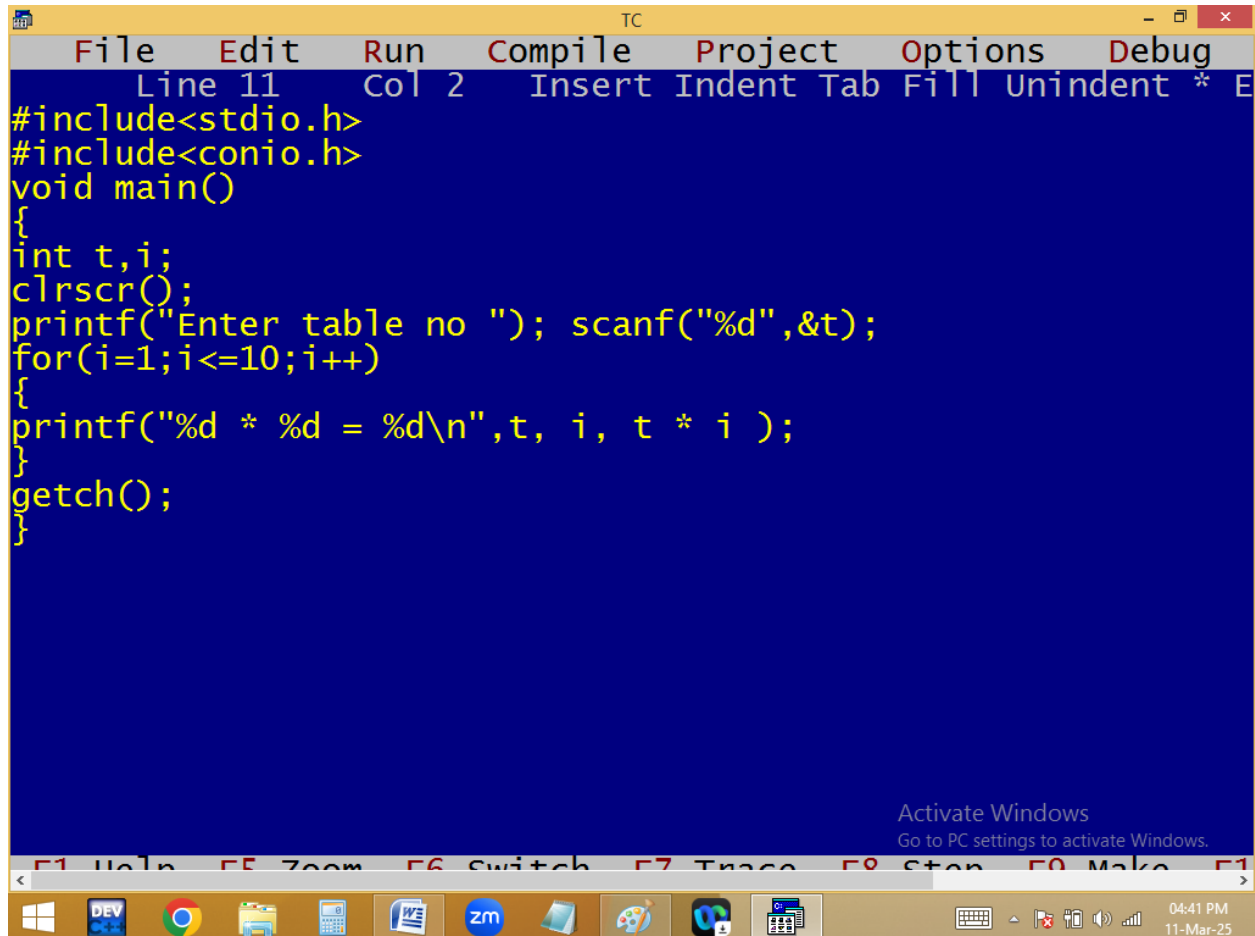
Eg: 7th table

7*1=7

7*2=14

$$7*3=21$$

$$7*10=70$$



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 11", "Col 2", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background with yellow text. The code is as follows:

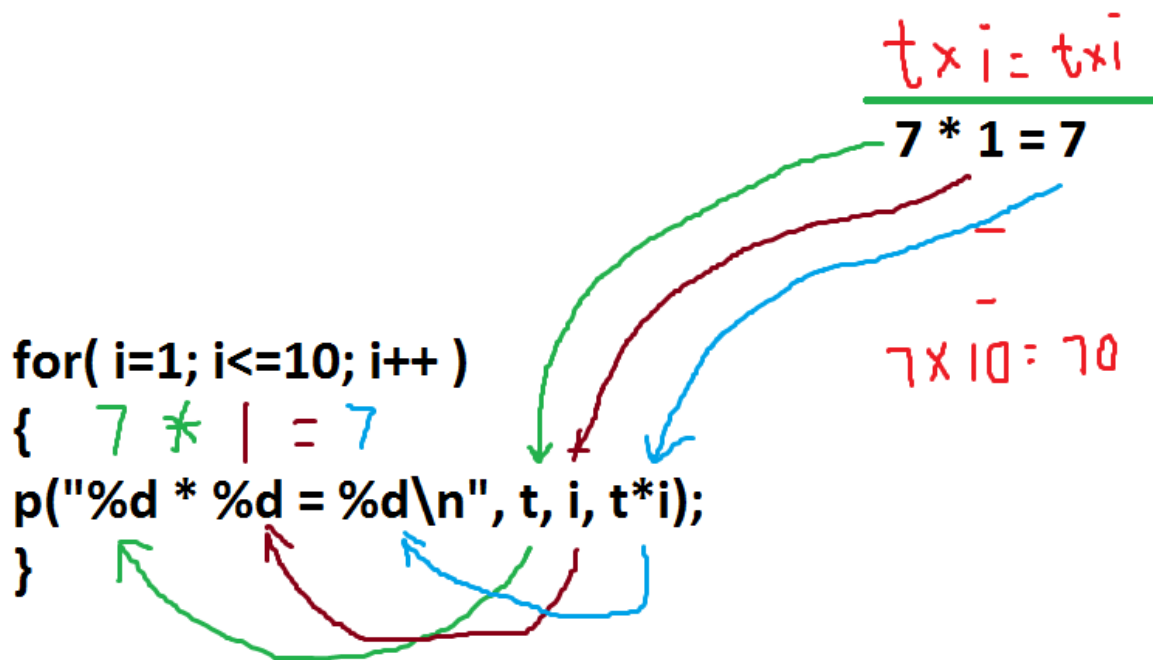
```
#include<stdio.h>
#include<conio.h>
void main()
{
int t,i;
clrscr();
printf("Enter table no "); scanf("%d",&t);
for(i=1;i<=10;i++)
{
printf("%d * %d = %d\n",t, i, t * i );
}
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, Chrome, File Explorer, Calculator, Word, Zoom, and others. On the right side of the taskbar, the system clock shows "04:41 PM" and "11-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

```
TC
Enter table no 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
```

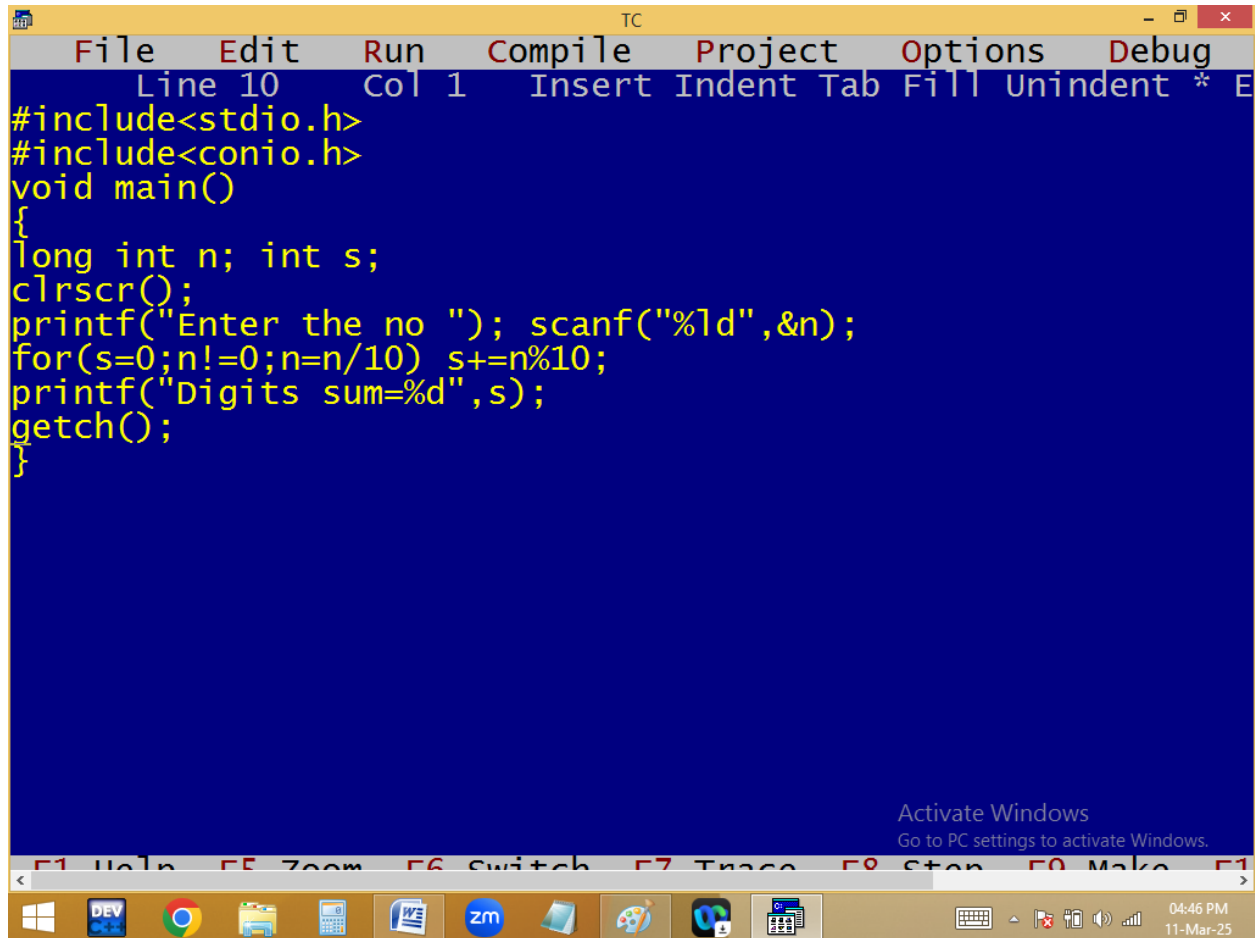
Activate Windows
Go to PC settings to activate Windows.

04:41 PM
11-Mar-25



Finding digits sum using for loop:

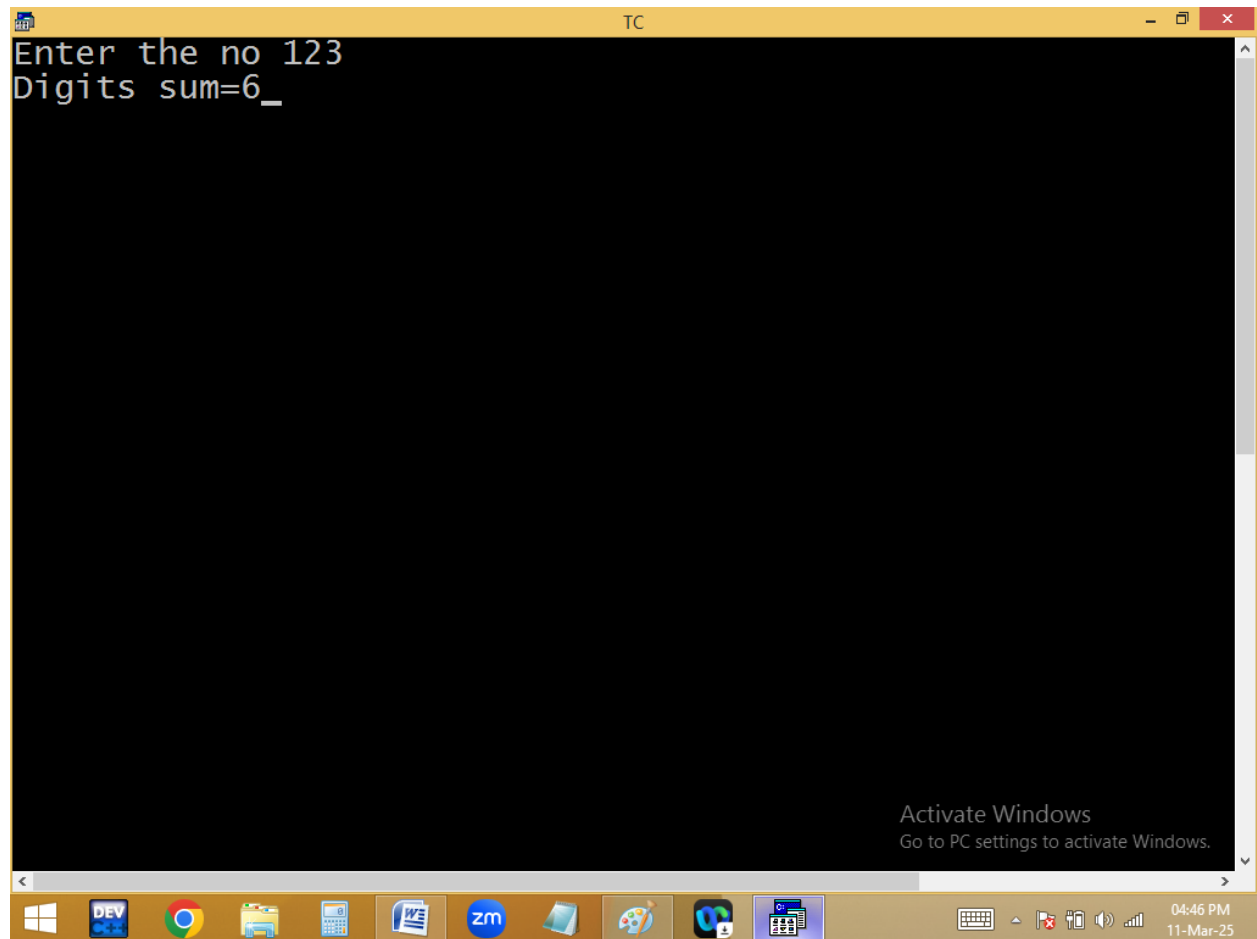
143 → 1+4+3=8

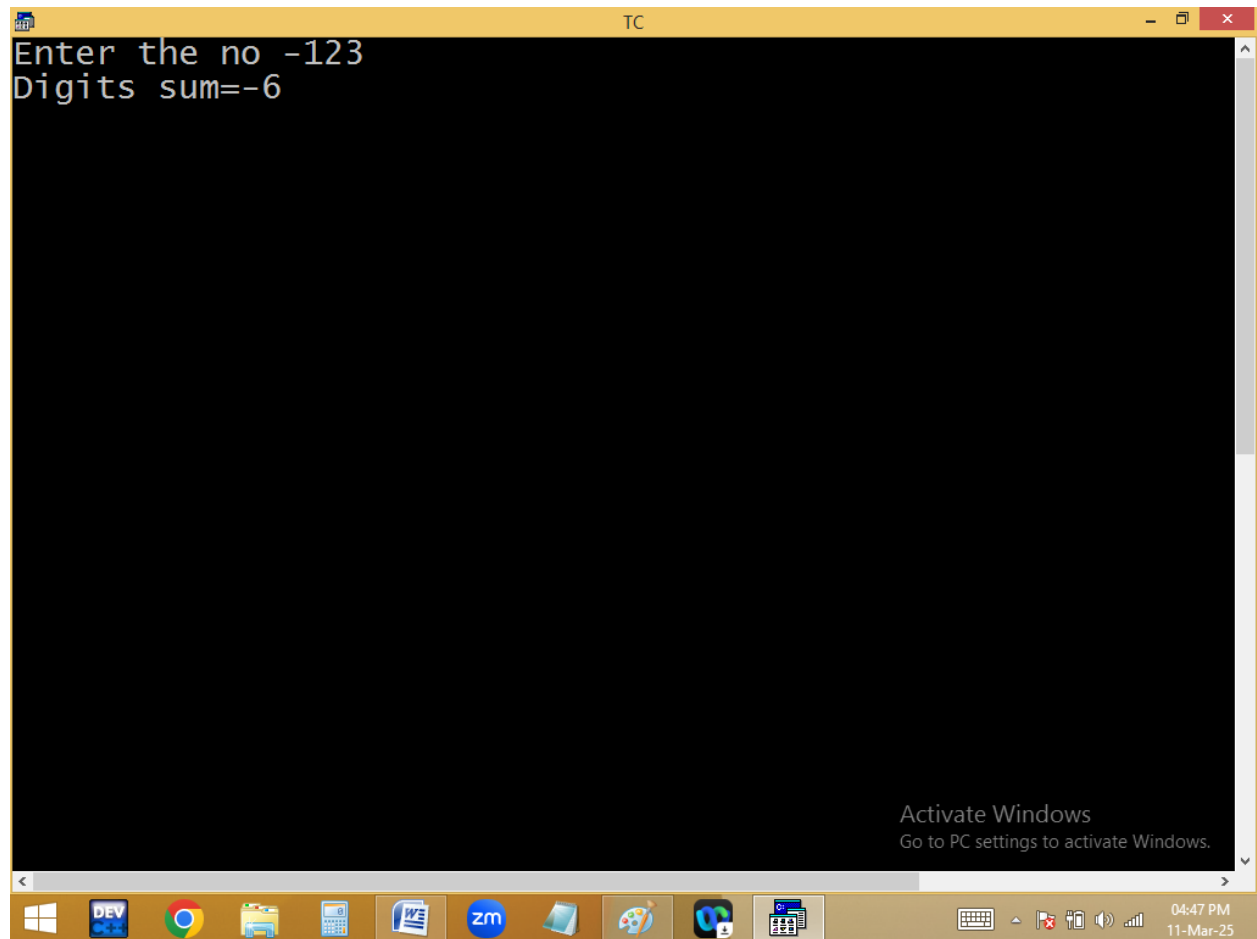


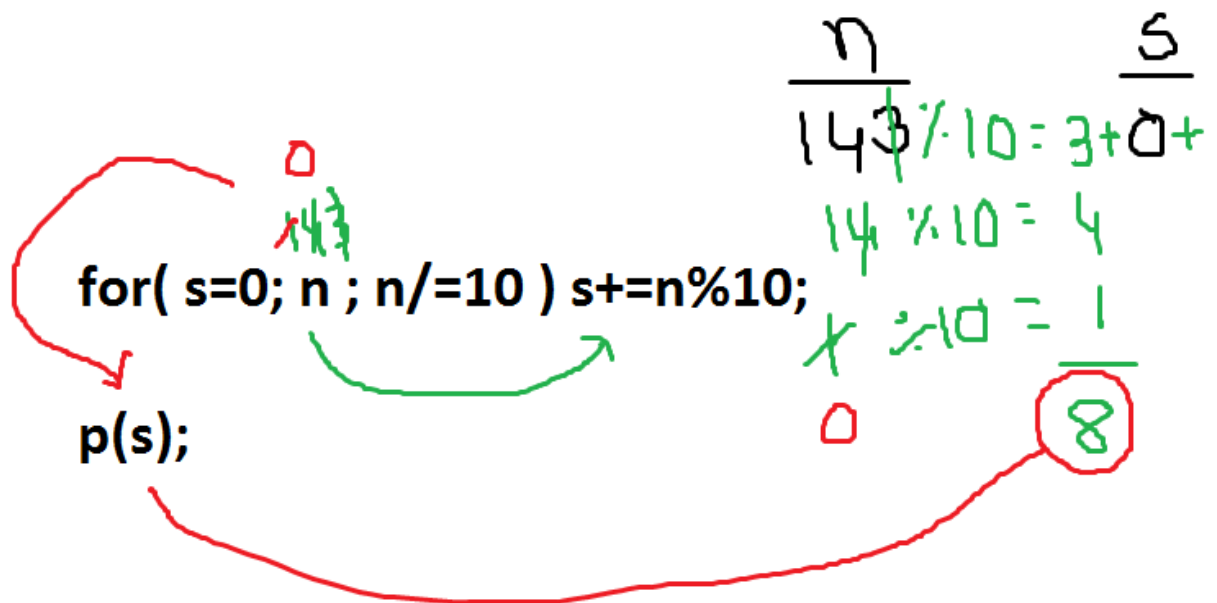
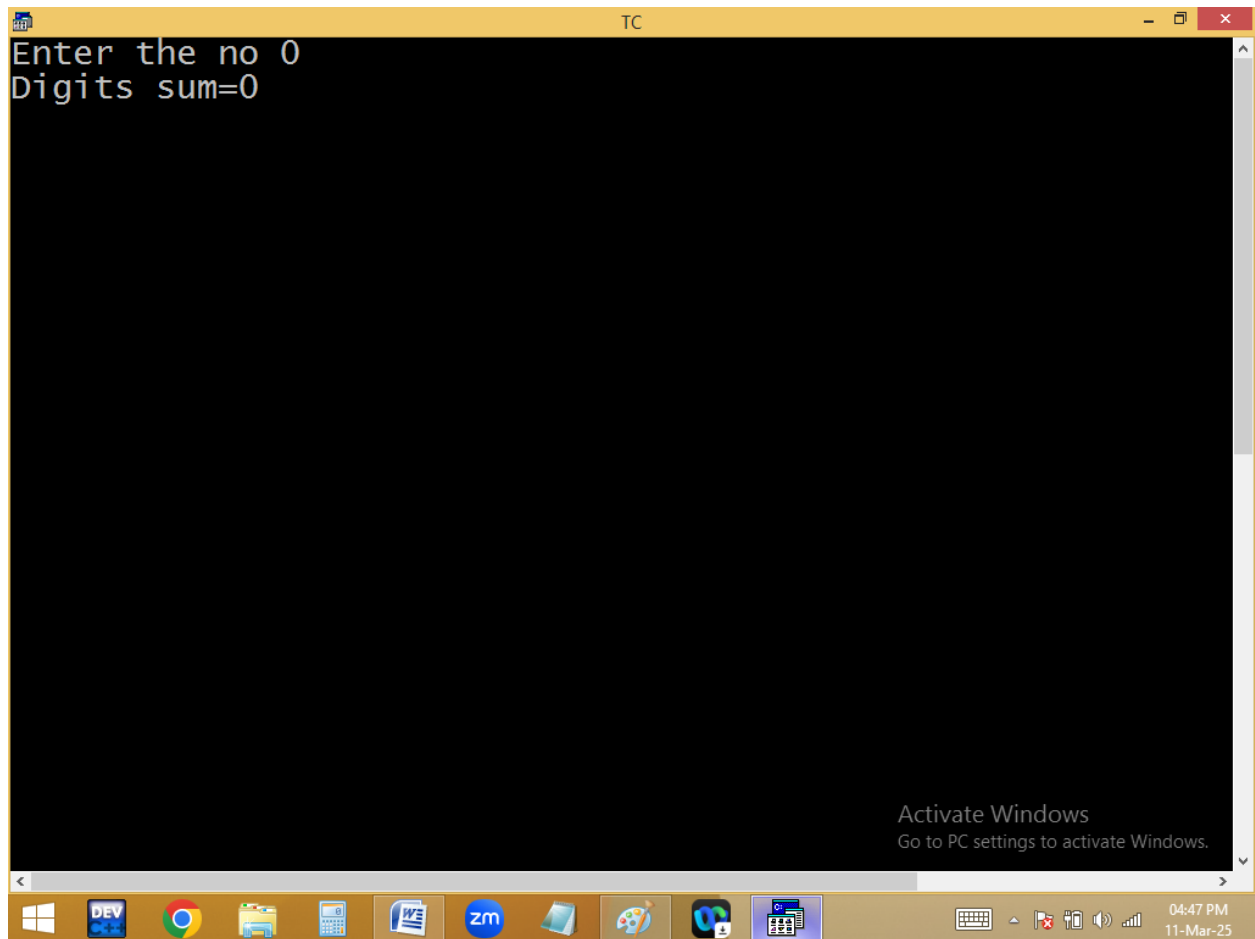
The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 10 Col 1'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long int n; int s;
    clrscr();
    printf("Enter the no "); scanf("%ld",&n);
    for(s=0;n!=0;n=n/10) s+=n%10;
    printf("Digits sum=%d",s);
    getch();
}
```

At the bottom of the window, there is an 'Activate Windows' watermark with the text 'Go to PC settings to activate Windows.' The Windows taskbar is visible at the very bottom, showing icons for DEV, Chrome, File Explorer, Calculator, Word, Zoom, and other applications. The system clock in the bottom right corner shows '04:46 PM 11-Mar-25'.

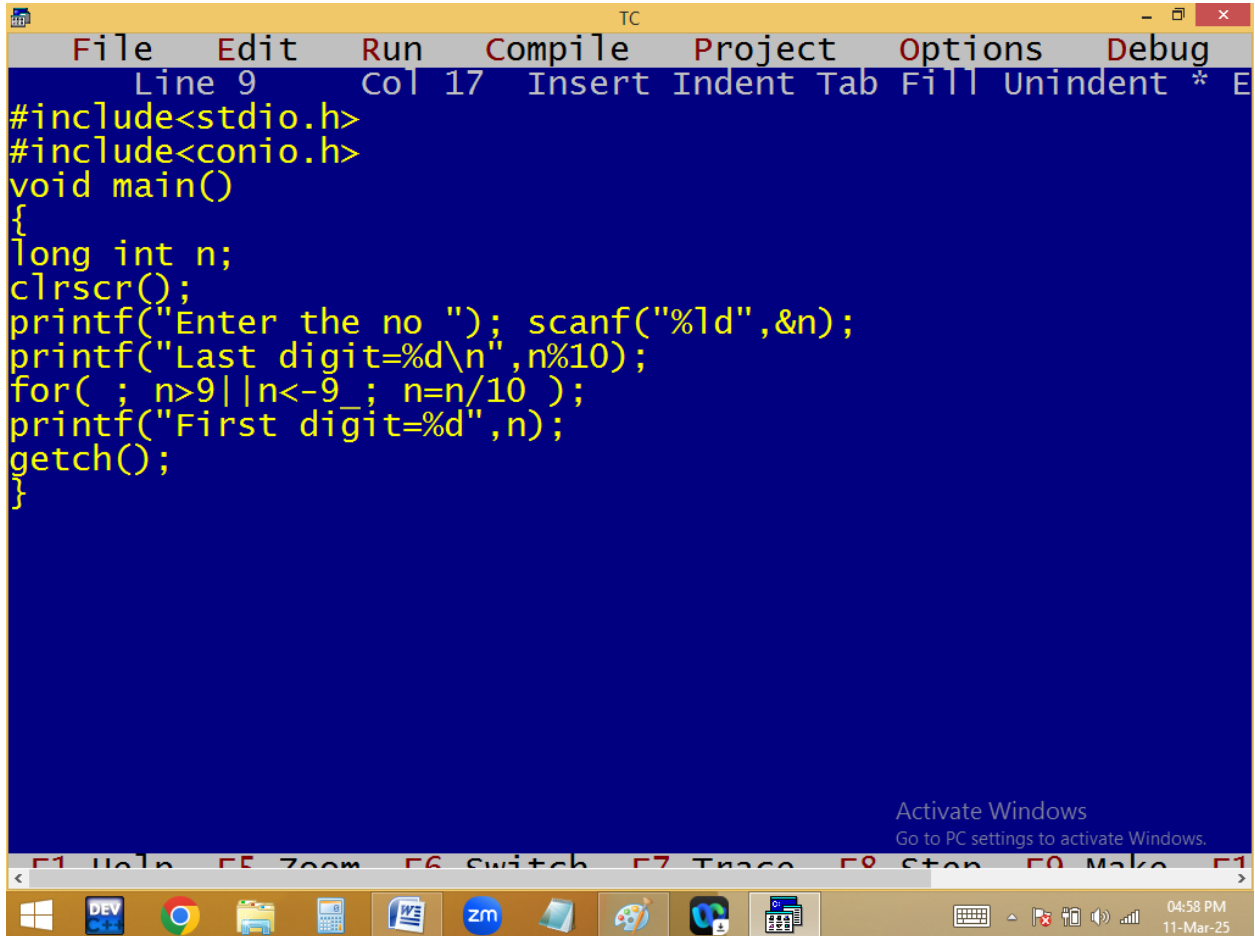






Printing last digit and first digit of given no?

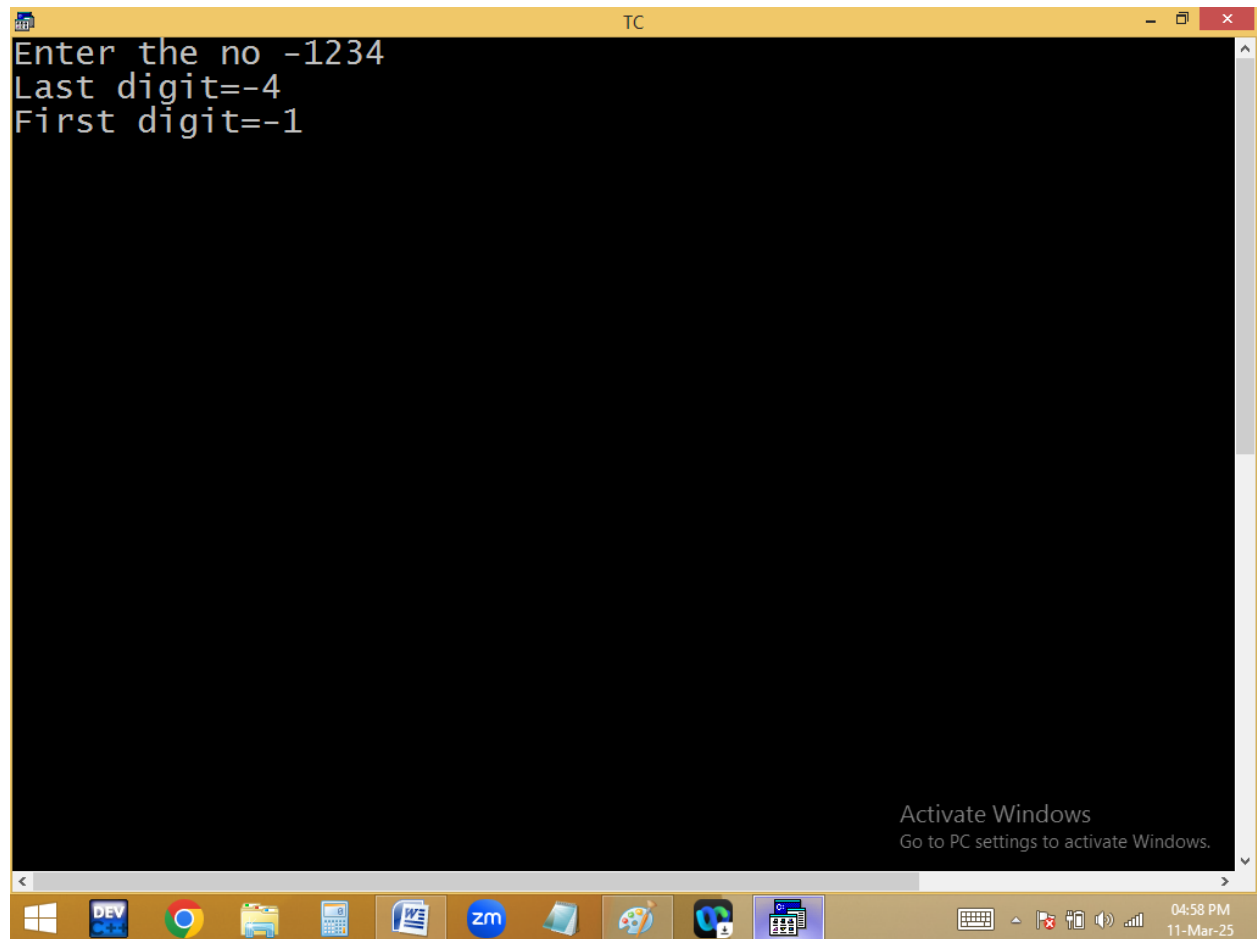
Eg: 2017 → 7 last digit, 2 first digit

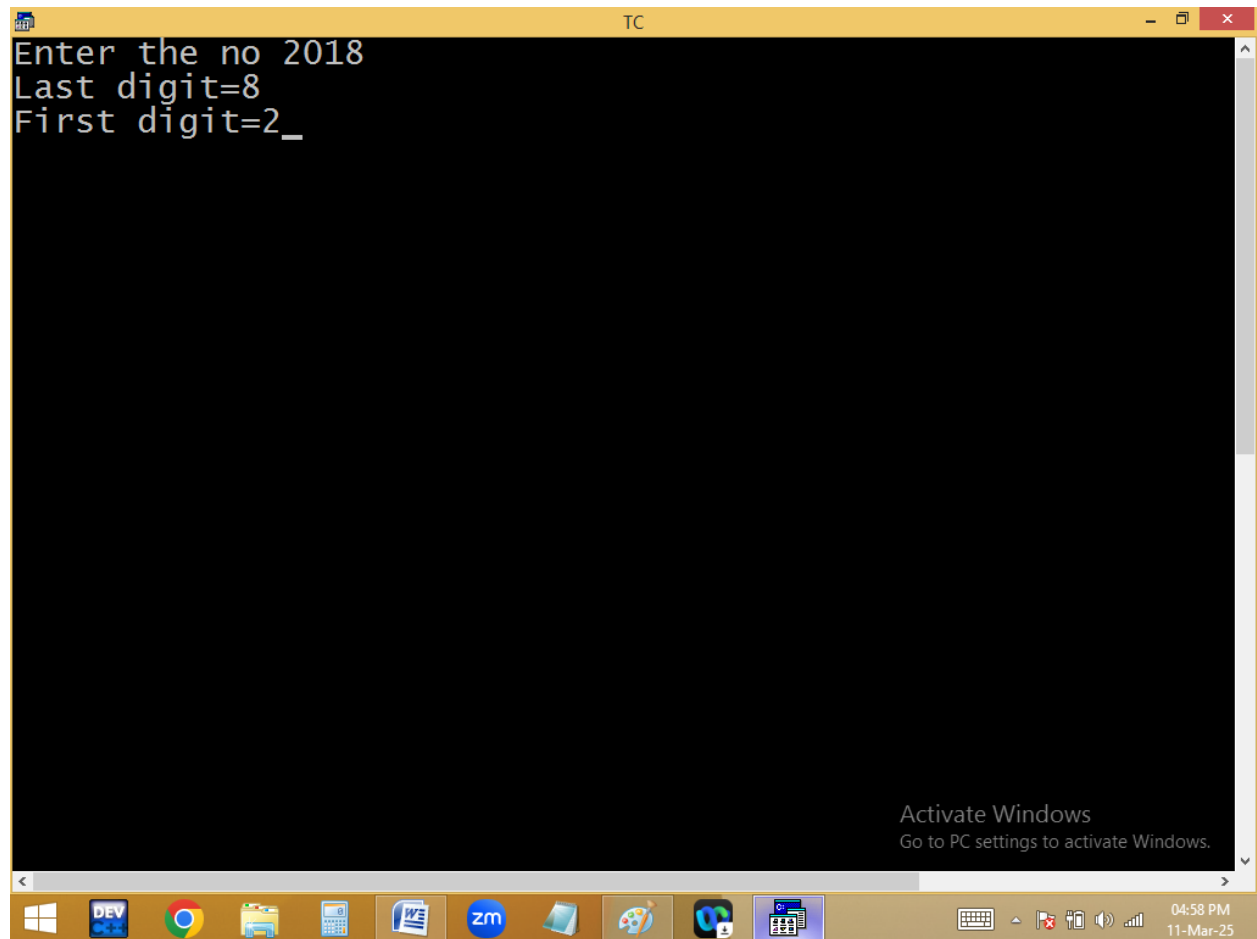


```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 17 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
long int n;
clrscr();
printf("Enter the no "); scanf("%ld",&n);
printf("Last digit=%d\n",n%10);
for( ; n>9||n<-9 ; n=n/10 );
printf("First digit=%d",n);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

04:58 PM
11-Mar-25





```
Enter the no 0
Last digit=0
First digit=0
```

Activate Windows
Go to PC settings to activate Windows.

7
p("Last digit=%d\n", n%10);

2017
for(; n>9 ; n/10)
{ }
p("First digit = %d",n);

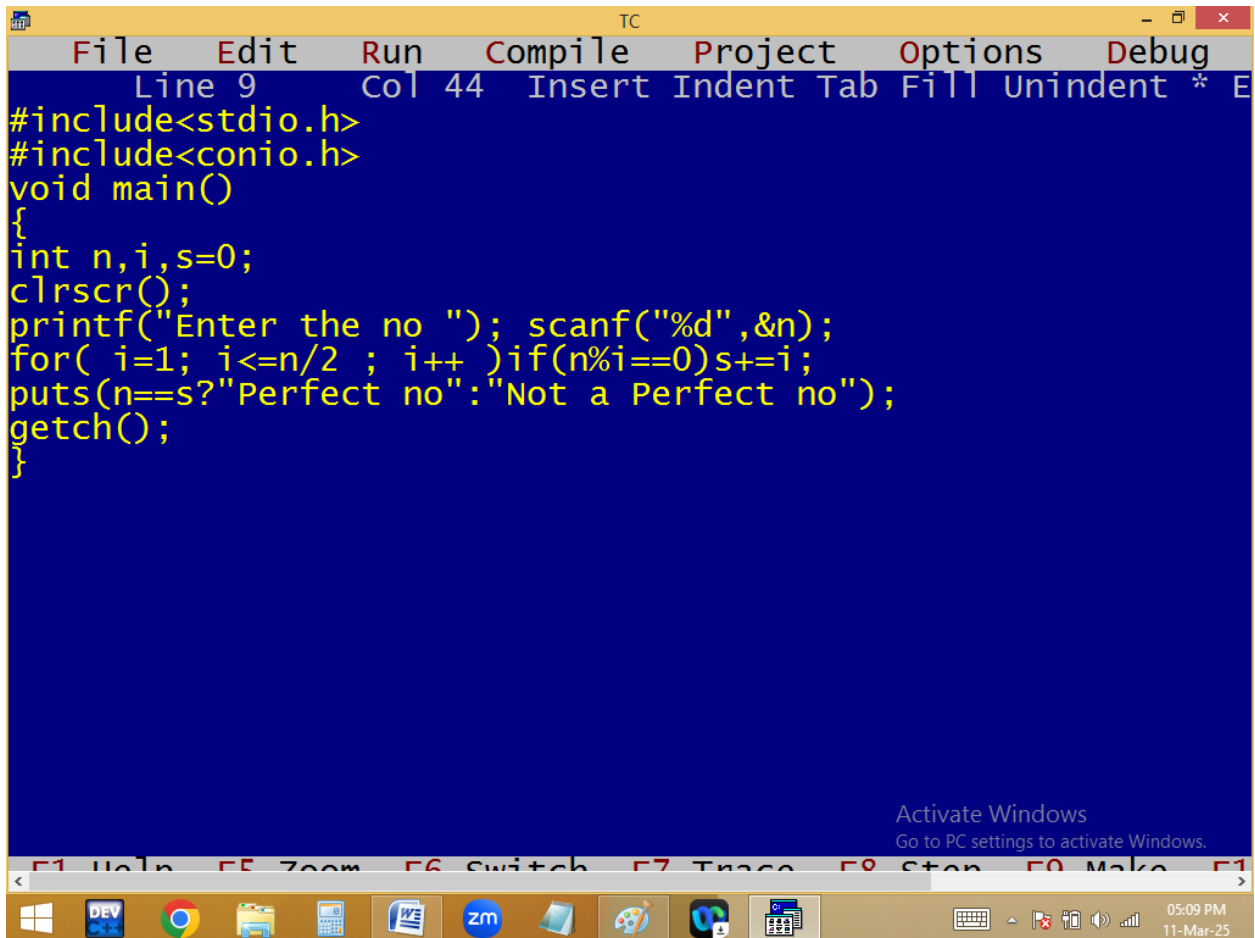
$$\frac{2017}{10} = 201.7$$

Finding perfect no or not?

Sum of factors is equal to given no

6 factors are $1 + 2 + 3 = 6$

28 factors are $1+2+4+7+14=28$



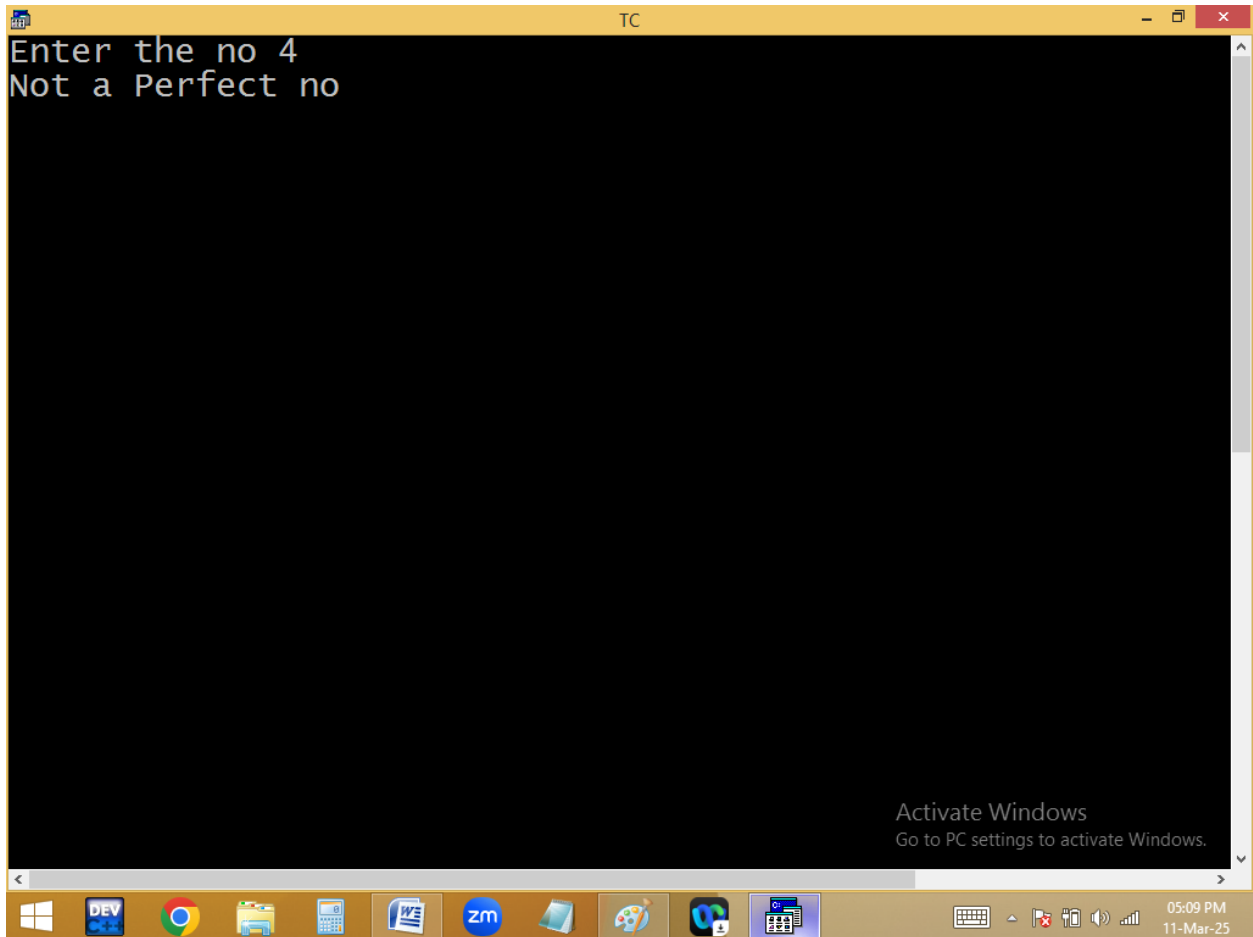
The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 9 Col 44' and provides keyboard shortcuts for Insert, Indent, Tab, Fill, Unindent, and End of Line (* E). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,i,s=0;
    clrscr();
    printf("Enter the no "); scanf("%d",&n);
    for( i=1; i<=n/2 ; i++ )if(n%i==0)s+=i;
    puts(n==s?"Perfect no":"Not a Perfect no");
    getch();
}
```

At the bottom of the window, there is a message: 'Activate Windows Go to PC settings to activate Windows.' The Windows taskbar is visible at the very bottom, showing icons for DEV, Chrome, File Explorer, Task Manager, Word, Zoom, and other applications. The system clock in the bottom right corner shows '05:09 PM 11-Mar-25'.







28/2
6/2

for(i=1;i<=n/2;i++) if(n%i==0) s+=i;

puts(n==s?"Perfect":"Not");

Handwritten notes:

Green:

n	i	s
6	1:0	0+1
6	2=0	2
6	3=0	3
		6

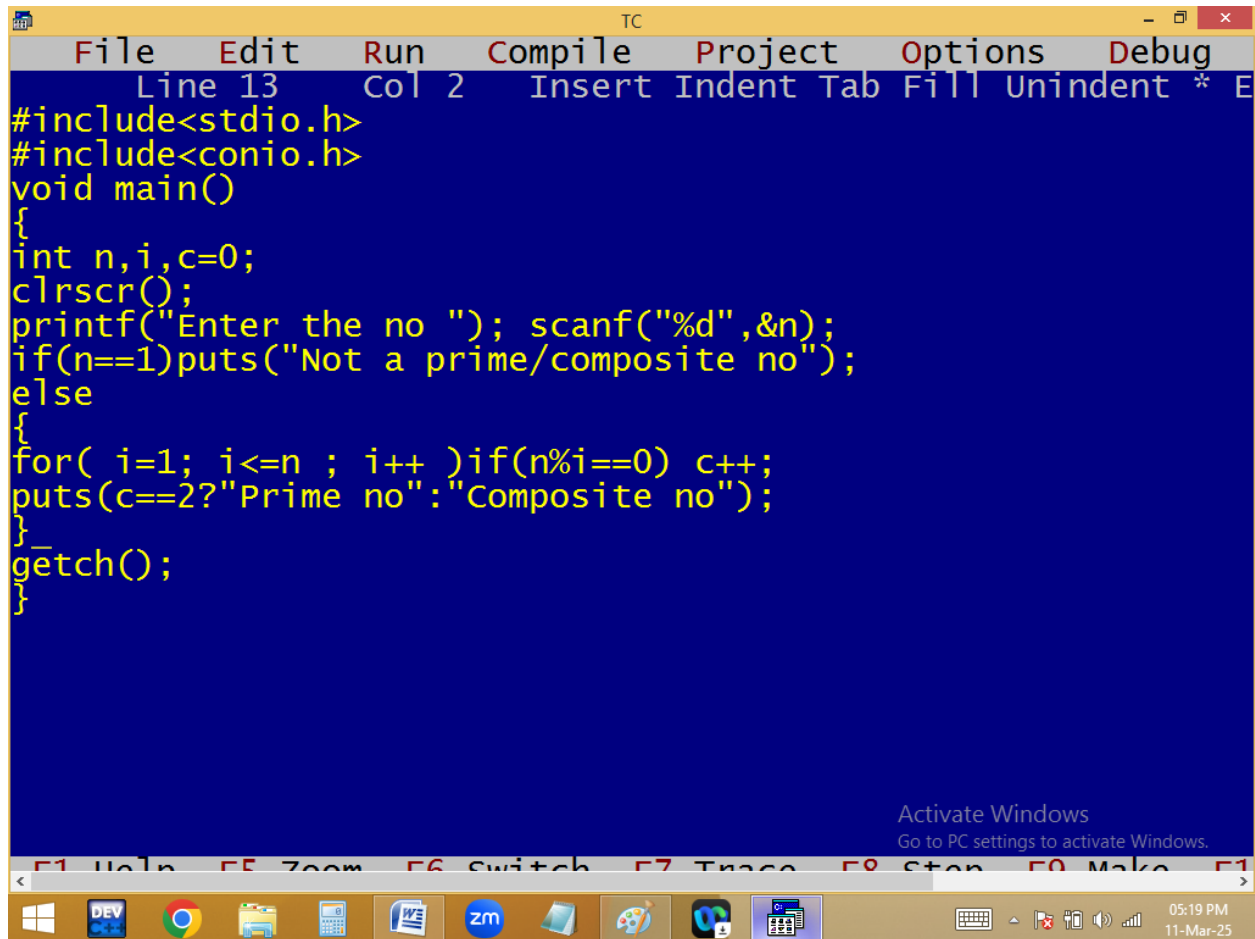
$4 \leq 3$

Blue:

4	1=0	1
4	2=0	2
		3

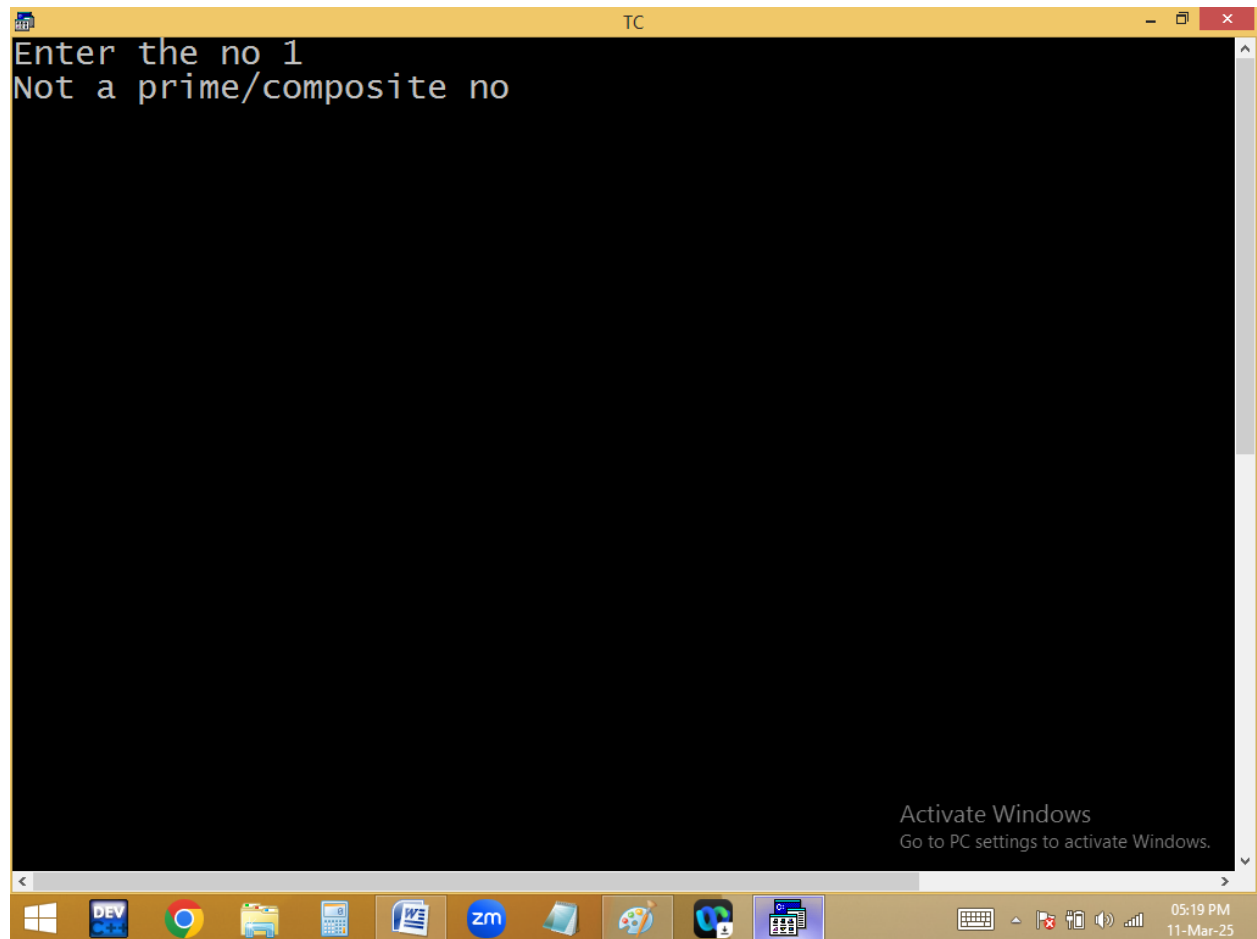
Red arrows indicate the flow of execution and the state of variables n, i, and s during the loop iterations.

Finding prime/composite no?



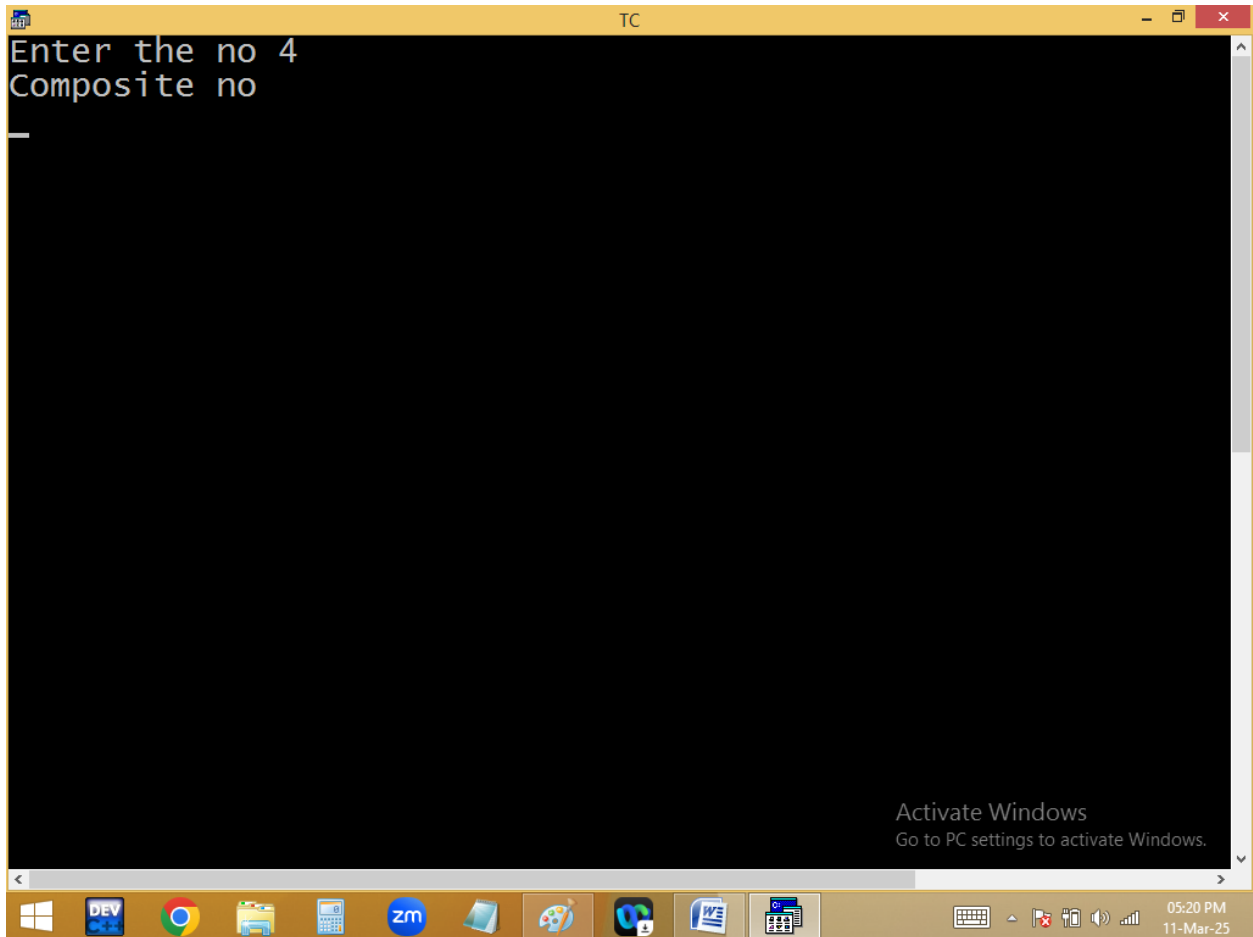
The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 13 Col 2' and lists editing actions: Insert, Indent, Tab, Fill, Unindent, and a wildcard * followed by E. The code is written in yellow text on a dark blue background. It includes headers for stdio.h and conio.h, and defines a main function. The program prompts the user to enter a number, checks if it's 1 (not prime/composite), and otherwise uses a loop to check for divisibility. If the number is divisible by any integer from 1 to n, it's composite; otherwise, it's prime. The program ends with a getch() call. The Windows taskbar at the bottom shows various icons including DEV, Chrome, File Explorer, Calculator, Word, Zoom, and others. The system clock shows 05:19 PM on 11-Mar-25. An 'Activate Windows' watermark is visible in the bottom right corner of the IDE window.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,c=0;
clrscr();
printf("Enter the no "); scanf("%d",&n);
if(n==1)puts("Not a prime/composite no");
else
{
for( i=1; i<=n ; i++ )if(n%i==0) c++;
puts(c==2?"Prime no":"Composite no");
}
getch();
}
```









no divisible with 1 and itself is prime

or

no having 2 factors only is called prime

$$1 \div 1 = 0$$

$$1 \div 1 = 0$$

n i

$$2 \% 1 = 0$$

$$2 \% 2 = 0$$

$$3 \% 1 = 0$$

$$3 \% 2 = 1$$

$$3 \% 3 = 0$$

$$4 \% 1 = 0$$

$$4 \% 2 = 0$$

$$4 \% 4 = 0$$

c

0

2 ✓

0

2 ✓

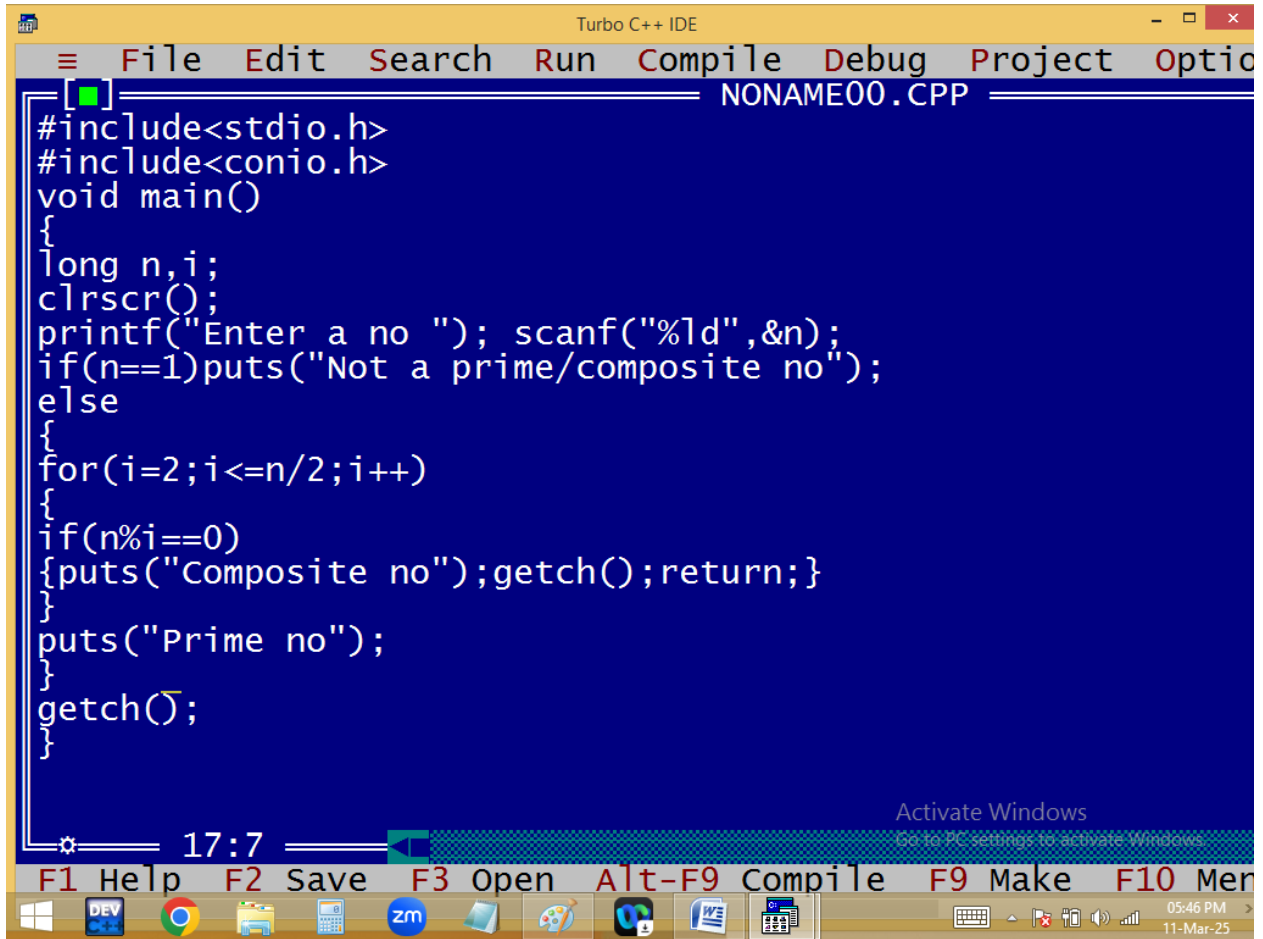
0

1

2

3 ✓

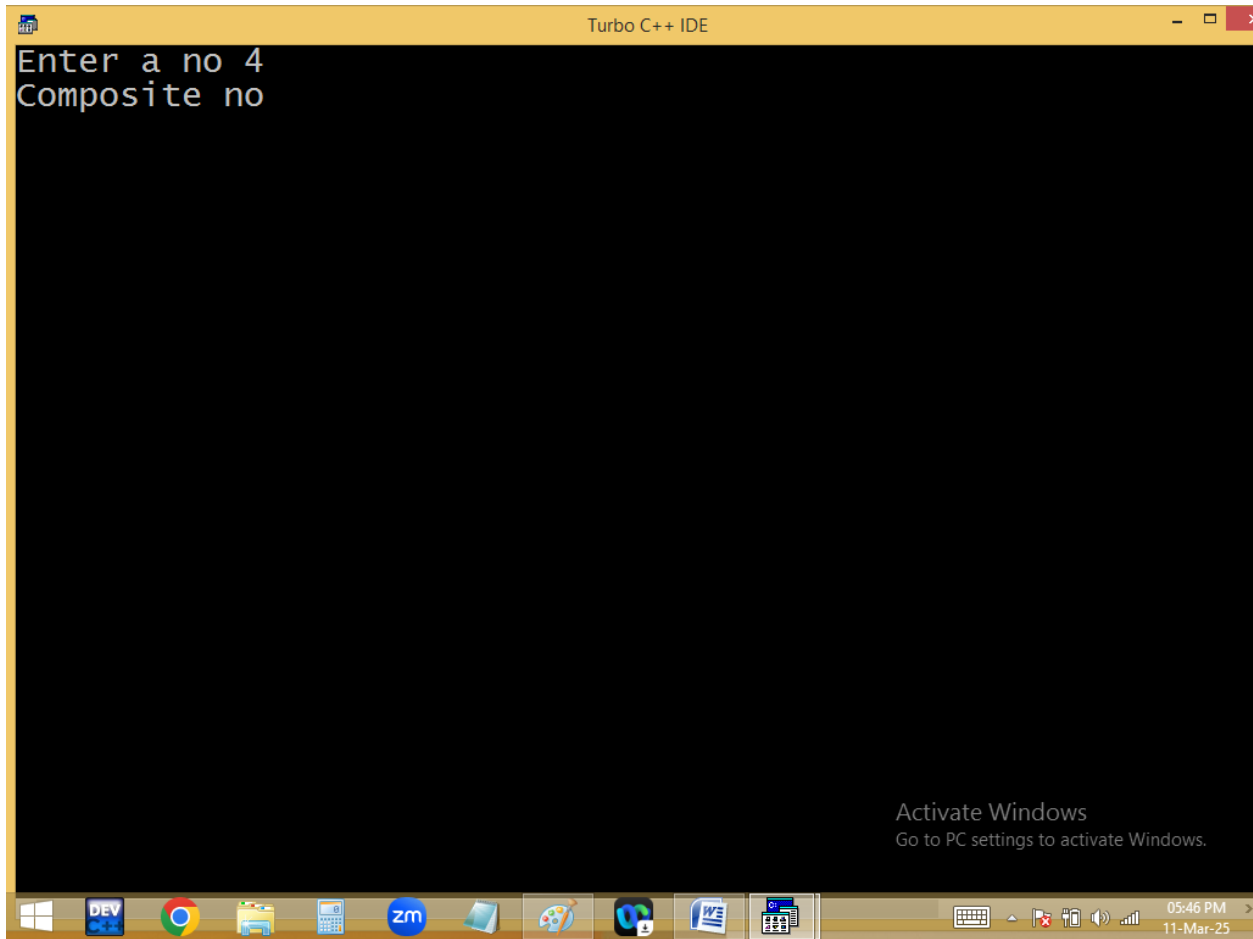
Method 2:

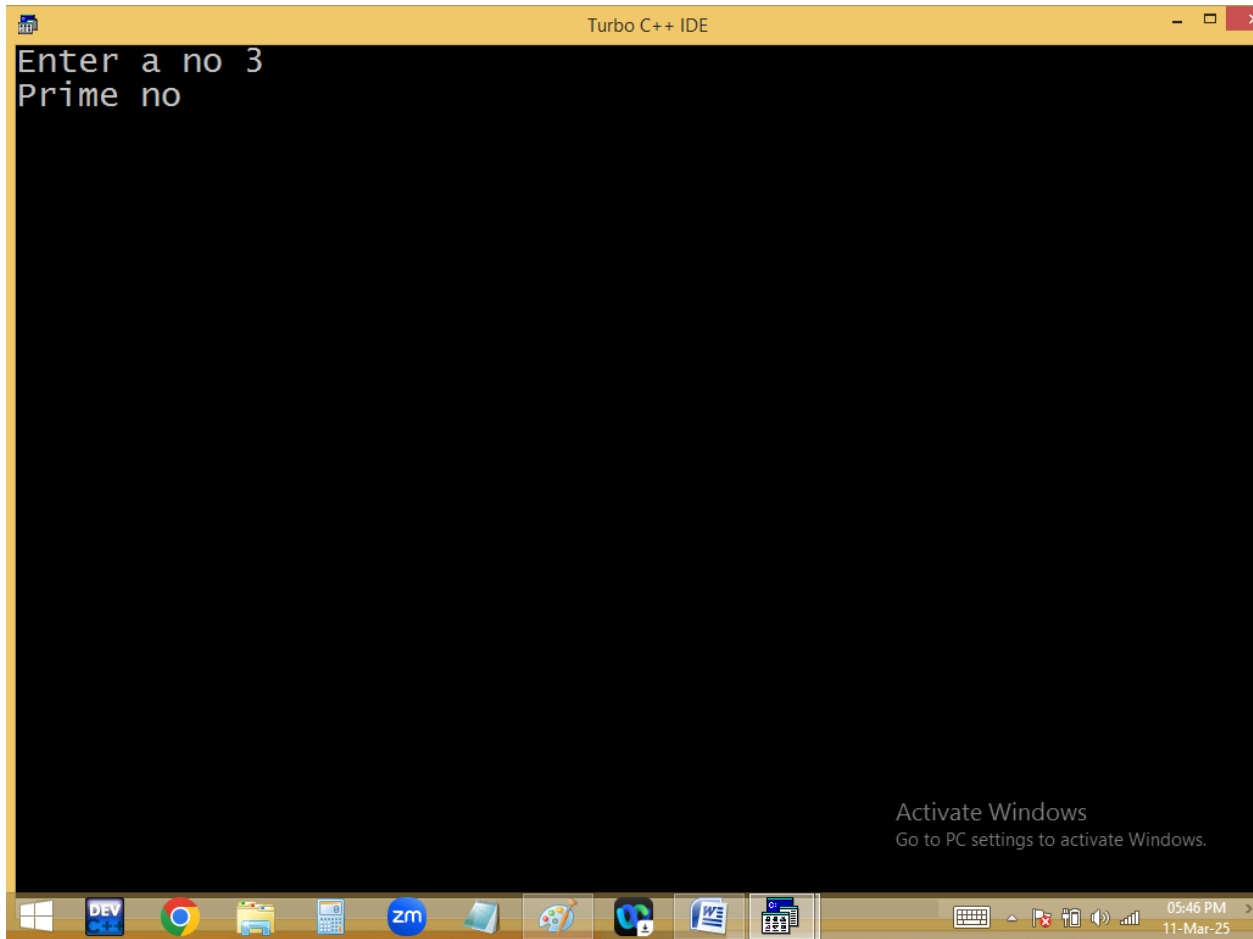


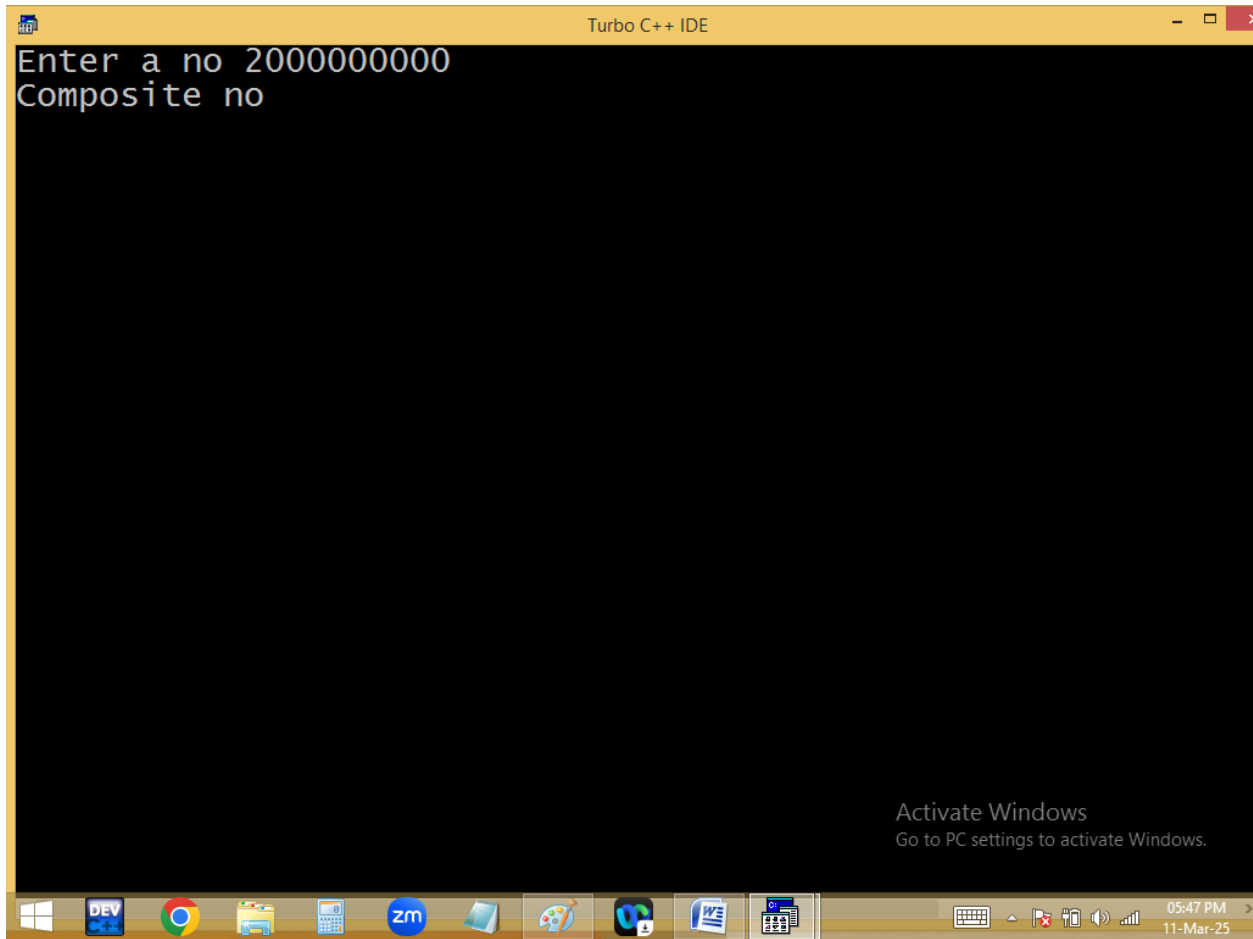
The screenshot shows the Turbo C++ IDE interface. The title bar reads "Turbo C++ IDE". The menu bar includes "File", "Edit", "Search", "Run", "Compile", "Debug", "Project", and "Options". The file name "NONAME00.CPP" is displayed in the top right. The code editor contains the following C++ code:

```
[■]
#include<stdio.h>
#include<conio.h>
void main()
{
    long n,i;
    clrscr();
    printf("Enter a no "); scanf("%ld",&n);
    if(n==1)puts("Not a prime/composite no");
    else
    {
        for(i=2;i<=n/2;i++)
        {
            if(n%i==0)
            {puts("Composite no");getch();return;}
        }
        puts("Prime no");
    }
    getch();
}
```

The status bar at the bottom shows the cursor position "17:7" and a "Go to PC settings to activate Windows" watermark. The function key bar includes "F1 Help", "F2 Save", "F3 Open", "Alt-F9 Compile", "F9 Make", and "F10 Men". The Windows taskbar at the very bottom shows various application icons and the system clock "05:46 PM 11-Mar-25".







no divisible with 1 and itself is prime

or

no having 2 factors only is called prime

$$1 \nmid 7, 1 = 0$$

$$1 \nmid 1, 1 = 0$$

10 - ~~1~~ 2 5 ~~10~~

100 - ~~1~~ 2 4 5 10 20 25 50 ~~100~~

11 - ~~1~~ ~~11~~

1. Fibonacci series:

n=5 → 0 1 1 2 3

2. printing below series:

n=5 → 1+2+3+4+5=15

3. printing below series:

$$n=5 \rightarrow 1-2+3-4+5=3$$

4.printing below series:

$$n=5 \rightarrow 1+1/1 + 1/2+1/3+1/4+1/5=3.28$$

5.printing below series:

$$n=5 \rightarrow 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$$

